

Please complete all of the following problems.
Show all your work.

5.8 Convert the following:

- (a) 0.745 atm to mmHg (b) 992 torr to bar
(c) 365 kPa to atm (d) 804 mmHg to kPa

5.18 Boyle's law relates gas volume to pressure, and Avogadro's law relates gas volume to number of moles. State a relationship between gas pressure and number of moles.

5.20 What is the effect of the following on the volume of 1 mol of an ideal gas?

- (a) The pressure is tripled (at constant T).
(b) The absolute temperature is increased by a factor of 2.5 (at constant P).
(c) Two more moles of the gas are added (at constant P and T).

5.22 What is the effect of the following on the volume of 1 mol of an ideal gas?

- (a) The temperature is decreased from 700 K to 350 K (at constant P).
(b) The temperature is increased from 350°C to 700°C (at constant P).
(c) The pressure is increased from 2 atm to 8 atm (at constant T).

5.24 A sample of sulfur hexafluoride gas occupies a volume of 5.10 L at 198°C. Assuming that the pressure remains constant, what temperature (in °C) is needed to reduce the volume to 2.50 L?

5.26 A sample of Freon-12 (CF_2Cl_2) occupies 25.5 L at 298 K and 153.3 kPa. Find its volume at STP.

5.28 A sample of chlorine gas is confined in a 5.0-L container at 228 torr and 27°C. How many moles of gas are in the sample?

5.30 You have 207 mL of chlorine trifluoride gas at 699 mmHg and 45°C. What is the mass (in g) of the sample?

5.33 A gas-filled weather balloon with a volume of 55.0 L is released at sea-level conditions of 755 torr and 23°C. The balloon can expand to a maximum volume of 835 L. When the balloon rises to an altitude at which the temperature is -5°C and the pressure is 0.066 atm, will it reach its maximum volume?

5.35 To collect a beaker of H_2 gas by displacing the air already in the beaker, would you hold the beaker upright or inverted? Why? How would you hold the beaker to collect CO_2 ?

5.38 What is the density of Xe gas at STP?

5.40 How many moles of gaseous arsine (AsH_3) will occupy 0.0400 L at STP? What is the density of gaseous arsine?

5.42 Calculate the molar mass of a gas at 388 torr and 45°C if 206 ng occupies 0.206 μL .

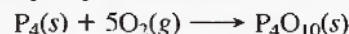
5.44 When 0.600 L of Ar at 1.20 atm and 227°C is mixed with 0.200 L of O_2 at 501 torr and 127°C in a 400-mL flask at 27°C, what is the pressure in the flask?

5.48 A sample of a liquid hydrocarbon known to consist of molecules with five carbon atoms is vaporized in a 0.204-L flask by immersion in a water bath at 101°C. The barometric pressure is 767 torr, and the remaining gas condenses to 0.482 g of liquid. What is the molecular formula of the hydrocarbon?

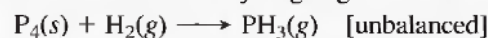
5.50 An environmental chemist is sampling industrial exhaust gases from a coal-burning plant. He collects a CO_2 - SO_2 - H_2O mixture in a 21-L steel tank until the pressure reaches 850. torr at 45°C.

- (a) How many moles of gas are collected?
(b) If the SO_2 concentration in the mixture is 7.95×10^3 parts per million by volume (ppmv), what is its partial pressure? [Hint: ppmv = (volume of component/volume of mixture) $\times 10^6$.]

5.51 How many grams of phosphorus react with 35.5 L of O_2 at STP to form tetraphosphorus decaoxide?



5.53 How many grams of phosphine (PH_3) can form when 37.5 g of phosphorus and 83.0 L of hydrogen gas react at STP?



5.55 Aluminum reacts with excess hydrochloric acid to form aqueous aluminum chloride and 35.8 mL of hydrogen gas over water at 27°C and 751 mmHg. How many grams of aluminum reacted?

5.57 "Strike anywhere" matches contain the compound tetraphosphorus trisulfide, which burns to form tetraphosphorus decaoxide and sulfur dioxide gas. How many milliliters of sulfur dioxide, measured at 725 torr and 32°C, can be produced from burning 0.800 g of tetraphosphorus trisulfide?

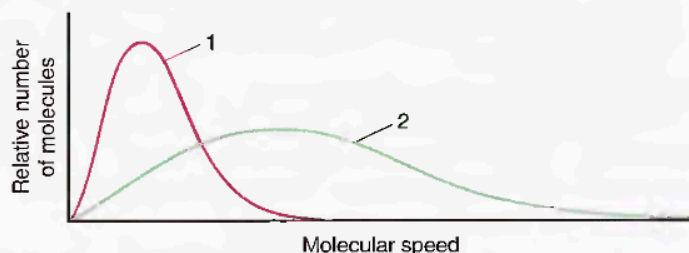
5.59 Xenon hexafluoride was one of the first noble gas compounds synthesized. The solid reacts rapidly with the silicon dioxide in glass or quartz containers to form liquid XeOF_4 and gaseous silicon tetrafluoride. What is the pressure in a 1.00-L container at 25°C after 2.00 g of xenon hexafluoride reacts? (Assume that silicon tetrafluoride is the only gas present and that it occupies the entire volume.)

5.63 How does the kinetic-molecular theory explain why 1 mol of krypton and 1 mol of helium have the same volume at STP?

5.66 Three 5-L flasks, fixed with pressure gauges and small valves, each contain 4 g of gas at 273 K. Flask A contains H_2 , flask B contains He, and flask C contains CH_4 . Rank the flask contents in terms of (a) pressure, (b) average molecular kinetic energy, (c) diffusion rate after the valve is opened, (d) total kinetic energy of the molecules, (e) density, and (f) collision frequency.

5.67 What is the ratio of effusion rates for the lightest gas, H_2 , and the heaviest known gas, UF_6 ?

5.69 The graph below shows the distribution of molecular speeds for argon and helium at the same temperature.



- Does curve 1 or 2 better represent the behavior of argon?
- Which curve represents the gas that effuses more slowly?
- Which curve more closely represents the behavior of fluorine gas? Explain.

5.71 At a given pressure and temperature, it takes 4.55 min for a 1.5-L sample of He to effuse through a membrane. How long does it take for 1.5 L of F_2 to effuse under the same conditions?

5.73 Solid white phosphorus melts and then vaporizes at high temperature. Gaseous white phosphorus effuses at a rate that is 0.404 times that of neon in the same apparatus under the same conditions. How many atoms are in a molecule of gaseous white phosphorus?

5.77 Does N_2 behave more ideally at 1 atm or at 500 atm? Explain.

5.80 Hemoglobin is the protein that transports O_2 through the blood from the lungs to the rest of the body. In doing so, each molecule of hemoglobin combines with four molecules of O_2 . If 1.00 g of hemoglobin combines with 1.53 mL of O_2 at $37^\circ C$ and 743 torr, what is the molar mass of hemoglobin?

5.83 Chlorine is produced from concentrated seawater by the electrochemical chlor-alkali process. During the process, the chlorine is collected in a container that is isolated from the other products to prevent unwanted (and explosive) reactions. If a 15.00-L container holds 0.5850 kg of Cl_2 gas at $225^\circ C$, calculate

- P_{IGL}
- P_{VDW} (use $R = 0.08206 \frac{\text{atm}\cdot\text{L}}{\text{mol}\cdot\text{K}}$)

(IGL: Ideal-Gas Law / VDW: Van der Waals)

5.86 When air is inhaled, it enters the alveoli of the lungs, and varying amounts of the component gases exchange with dissolved gases in the blood. The resulting alveolar gas mixture is quite different from the atmospheric mixture. The following table presents selected data on the composition and partial pressure of four gases in the atmosphere and in the alveoli:

Gas	Atmosphere (sea level)		Alveoli	
	Mol %	Partial Pressure (torr)	Mol %	Partial Pressure (torr)
N_2	78.6	—	—	569
O_2	20.9	—	—	104
CO_2	0.04	—	—	40
H_2O	0.46	—	—	47

If the total pressure of each gas mixture is 1.00 atm, calculate the following:

- The partial pressure (in torr) of each gas in the atmosphere
- The mol % of each gas in the alveoli
- The number of O_2 molecules in 0.50 L of alveolar air (volume of an average breath at rest) at $37^\circ C$

5.88 At 1400. mmHg and 286 K, a skin diver exhales a 208-mL bubble of air that is 77% N_2 , 17% O_2 , and 6.0% CO_2 by volume.

- How many milliliters would the volume of the bubble be if it were exhaled at the surface at 1 atm and 298 K?
- How many moles of N_2 are in the bubble?

5.95 Aluminum chloride is easily vaporized at temperatures above $180^\circ C$. The gas escapes through a pinhole 0.122 times as fast as helium at the same conditions of temperature and pressure in the same apparatus. What is the molecular formula of gaseous aluminum chloride?

5.97 An atmospheric chemist studying the reactions of the pollutant SO_2 places a mixture of SO_2 and O_2 in a 2.00-L container at 900. K and an initial pressure of 1.95 atm. When the reaction occurs, gaseous SO_3 forms, and the pressure eventually falls to 1.65 atm. How many moles of SO_3 form?

5.101 Canadian chemists have developed a modern variation of the 1899 Mond process for preparing extremely pure metallic nickel. A sample of impure nickel reacts with carbon monoxide at $50^\circ C$ to form gaseous nickel carbonyl, $Ni(CO)_4$.

- How many grams of nickel can be converted to the carbonyl with 3.55 m^3 of CO at 100.7 kPa?
- The carbonyl is then decomposed at 21 atm and $155^\circ C$ to pure ($>99.95\%$) nickel. How many grams of nickel are obtained per cubic meter of the carbonyl?
- The released carbon monoxide is cooled and collected for reuse by passing it through water at $35^\circ C$. If the barometric pressure is 769 torr, what volume (in m^3) of CO is formed per cubic meter of carbonyl?

5.103 A gaseous organic compound containing only carbon, hydrogen, and nitrogen is burned in oxygen gas, and the individual volume of each reactant and product is measured under the same conditions of temperature and pressure. Reaction of four volumes of the compound produces four volumes of CO_2 , two volumes of N_2 , and ten volumes of water vapor. (a) What volume of oxygen gas was required? (b) What is the empirical formula of the compound?

5.107 By what factor would a scuba diver's lungs expand if she ascended rapidly to the surface from a depth of 125 ft without inhaling or exhaling? If an expansion factor greater than 1.5 causes lung rupture, how far could she safely ascend from 125 ft without breathing? Assume constant temperature (d of seawater = 1.04 g/mL; d of Hg = 13.5 g/mL).

5.109 Dilute aqueous hydrogen peroxide is used as a bleaching agent and for disinfecting surfaces and small cuts. Its concentration is sometimes given as a certain number of "volumes hydrogen peroxide," which refers to the number of volumes of O_2 gas, measured at STP, that a given volume of hydrogen peroxide solution will release when it decomposes to O_2 and liquid H_2O . How many grams of hydrogen peroxide are in 0.100 L of "20 volumes hydrogen peroxide" solution?

5.121 A mixture of CO_2 and Kr weighs 35.0 g and exerts a pressure of 0.708 atm in its container. Since Kr is expensive, you wish to recover it from the mixture. After the CO_2 is completely removed by absorption with $NaOH(s)$, the pressure in the container is 0.250 atm. How many grams of CO_2 were originally present? How many grams of Kr can you recover?

5.126 The lunar surface reaches 370 K at midday. The atmosphere consists of neon, argon, and traces of helium at a total pressure of only 2×10^{-14} atm. Calculate the rms speed of each component in the lunar atmosphere. [Use $R = 8.314 \text{ J/(mol}\cdot\text{K)}$ and M in kg/mol.]

5.135 A large portion of metabolic energy arises from the biological combustion of glucose:



(a) If this reaction is carried out in an expandable container at 35°C and 780. torr, what volume of CO_2 is produced from 18.0 g of glucose and excess O_2 ? (b) If the reaction is carried out at the same conditions with the stoichiometric amount of O_2 , what is the partial pressure of each gas when the reaction is 50% complete (9.0 g of glucose remains)?

5.140 One way to utilize naturally occurring uranium (0.72% ^{235}U and 99.27% ^{238}U) as a nuclear fuel is to enrich it (increase its ^{235}U content) by allowing gaseous UF_6 to effuse through a porous membrane (see the margin note, p. 205). From the relative rates of effusion of $^{235}\text{UF}_6$ and $^{238}\text{UF}_6$, find the number of steps needed to produce uranium that is 3.0 mole % ^{235}U , the enriched fuel used in many nuclear reactors.

5.142 In preparation for a combustion demonstration, a professor's assistant fills a balloon with equal molar amounts of H_2 and O_2 , but the demonstration has to be postponed until the next day. During the night, both gases leak through pores in the balloon. If 45% of the H_2 leaks, what is the O_2/H_2 ratio in the balloon the next day?

5.146 When gaseous F_2 and solid I_2 are heated to high temperatures, the I_2 sublimates and gaseous iodine heptafluoride forms. If 350. torr of F_2 and 2.50 g of solid I_2 are put into a 2.50-L container at 250. K and the container is heated to 550. K, what is the final pressure? What is the partial pressure of I_2 gas?